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Class 11 Chemistry Notes Chapter 10 The s-block elements

Introduction:

In the previous chapter we have discussed about The Hydrogen but in this chapter we will study the general characteristics of the alkali and alkaline earth metals and their compounds. We will also study the compounds of s-block elements, their uses and importance, commercially and industrially. The biological significance of sodium, potassium, calcium and magnesium will also be discussed in this chapter.

The s-Block Elements

The s-block elements of the Periodic Table are those in which the last electron enters the outermost s-orbital. As the s-orbital can accommodate only two electrons, two groups (1 & 2) belong to the s-block of the Periodic Table.

Group-1 of periodic table contains

Lithium (Li), Sodium (Na), Potassium (K), Rubidium (Rb), Caesium (Cs) and Francium (Fr). Together these elements are called alkali metals because they form hydroxides on reaction with water, which are strongly alkaline in nature.

The group-2

Includes Beryllium (Be), Magnesium (Mg), Calcium (Ca), Strontium (Sr), Barium (Ba) and Radium (Ra). Except Beryllium, rest of the elements of group-2 are called the alkaline earth metals. These are called so because their oxides and hydroxides are alkaline in nature and these metal oxides are found in the earth crust.

Group-1 Elements: Alkali Metals

1. Electronic Configuration

Electronic Configuration of elements of group-1 is ns^1, where n represents the valence shell. The alkali metals have one valence electron, outside the noble gas core.

Element	Symbol	Electronic configuration
Lithium	Li	$1s^2 2s^1$
Sodium	Na	$1s^2 2s^2 2p^6 3s^1$
Potassium	K	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
Rubidium	Rb	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 5s^1$
Caesium	Cs	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6 6s^1$ or [Xe] $6s^1$
Francium	Fr	[Rn] $7s^1$

2. Atomic and ionic radii

The atoms of alkali metals have the largest size in their respective periods. The atomic radius increases on moving down the group because on moving down the group there is a progressive addition of new energy shells.

3. Ionization enthalpy

The ionization enthalpies of the alkali metals are generally low and decrease down the group from Li to Cs. This is because on moving down the group is due to increase in size of the atoms of alkali metals and increase in the magnitude of screening effect.

4. Hydration enthalpy

The alkali metal ions are extensively hydrated in aqueous solutions. The hydration enthalpies of alkali metal ions decrease with increase in ionic size $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$.

5. Physical properties

(i). Alkali metals are silvery white in colour and are generally soft and light metals.
(ii). The densities of alkali metals are low and increase down the group. Alkali metals have low melting and boiling point.
(iii). When alkali are heated metals they impart characteristic colours to the flame.
(Iv). When the excited electron comes back to the ground state, there is emission of radiation in the visible region.

6. Chemical Properties

The alkali metals are highly reactive elements. The cause for their high chemical reactivity is:
(i). Low value of first ionisation enthalpy
(ii). Large size
(iii). low heat of atomisation.

(i). Reaction with Air: Alkali metals burn very fast in oxygen and form different kind of oxides like monoxides, peroxides and superoxides. In all the compounds formed by alkali metals with oxygen, their oxidation state is +1.

$$4Li + O_2 \rightarrow 2Li_2O \text{ (Oxide)}$$
$$Na + O_2 \rightarrow Na_2O_2 \text{ (Peroxide)}$$
$$M + O_2 \rightarrow MO_2 \text{ (Superoxide)}$$

(ii). Reaction with Water: The alkali metals on reaction with water form their respective hydroxide and dihydrogen.

$$2M + 2H_2O \rightarrow 2M^+ + 2OH^- + H_2$$

(M = an alkali metal)

(iii). Reaction with Dihydrogen: Alkali metal react with dry di-hydrogen at about 673K (lithium at 1073K) to form crystalline hydrides which are ionic in nature and have high melting points.

$$2M + H_2 \rightarrow 2M^+H^-$$

(iv). Reaction with Halogens: The alkali metals react vigorously with halogens and form halides which are ionic in nature, M^+X^-. But the halides of lithium are a bit covalent in nature.

(v). Reaction with Mercury: The alkali metals have strong tendency to get oxidised, that is why they act as strong reducing agents, among these lithium is the strongest and sodium is the least powerful reducing agent.

(vi). Reducing Nature: Alkali metals combine with mercury to form amalgams. The reaction is highly exothermic in nature.

$$Na + Hg \rightarrow Na[Hg]$$

(vii). Solutions in liquid Ammonia: All alkali metals dissolve in liquid ammonia and give deep blue colour solution which are conducting in nature. These solutions contain ammoniated cations and ammoniated electrons as shown below:

$$M + (x + y)NH_3 \rightarrow [M(NH_3)x]+ + [e(NH_3)y]^-$$

Uses of Alkali Metals

1. Lithium is used as a metal in a number of alloys. Its alloys with aluminium to make aircraft parts.
2. Lithium hydroxides is used in the ventilation systems of space crafts and submarines to absorb carbondioxide.
3. Lithium aluminium hydride (LiAlH4) is a powerful reducing agent which is commonly used in organic synthesis.
4. Liquid sodium or its alloys with potassium is used as a coolant in nuclear reactors.
5. Sodium-lead alloy is used for the preparation of tetraethyl lead, Pb(C2H5)4, which is used as an antiknocking agent in petrol.
6. Sodium is used in the production of sodium vapour lamps.
7. Potassium chloride is used as fertilizer.
8. Potassium hydroxide is used in the manufacture of soft soaps and also as absorbent of carbon dioxide.
9. Potassium ions play a vital role in biological systems.
10. Caesium is used in photoelectric cells.

Anomalous Properties of Lithium

Lithium shows properties which are very different from the other members of its group. This is due to the:

1. Exceptionally small size of its atom and ion.
2. Greater polarizing power of lithium ion.
3. As compared to other alkali metals, lithium is harder and its melting point and boiling point are higher.
4. Among all the alkali metals lithium is least reactive but the strongest reducing agent.

Some important Compounds of Sodium

Sodium is highly reactive and always found in combined state. The isotope of sodium (Na) is used in detection of leukemia. The compound of sodium are given below:

1. Sodium Oxide (Na_2O)

Preparation:

$2NaNO_2 + 6Na \rightarrow 4Na_2O + N_2$

Properties:

1. Sodium oxide is a colourless ionic solid.
2. Aqueous solution of sodium oxide is strongly basic.
3. Sodium oxide on reaction with liquid ammonia forms sodamide.
4. At low temperature, when sodium peroxide is reacted with water or acids, H_2O_2 is formed.

2. Sodium Peroxide (Na_2O_2)

Preparation:

Sodium when heated in excess of air or when heated in excess of pure oxygen gives sodium peroxide.

$2Na + O_2 \rightarrow Na_2O_2$

Properties:

1. Sodium peroxide is a pale yellow diamagnetic compound.
2. Sodium peroxide is a powerful oxidising agent.
3. Sodium peroxide combines with CO and CO_2 to give carbonate.
4. At low temperature, when sodium peroxide is reacted with water or acids, H_2O_2 is formed.

$Na_2O_2 + 2H_2O \rightarrow 2NaOH + H_2O_2$

3. Sodium Hydroxide (Caustic Soda) (NaOH)

Preparation:

When sodium carbonate is treated with calcium hydroxide it give calcium carbonate along with sodium hydroxide. Also known as lime caustic soda process. It is a reversible reaction.

$Na_2CO_3 + Ca(OH)_2 \rightarrow 2NaOH + CaCO_3$

Properties

1. Sodium hydroxide is a white crystalline deliquescent solid.
2. Sodium hydroxide is corrosive in nature.
3. Sodium hydroxide is highly soluble in water.
4. Sodium hydroxide reacts with acid forming corresponding salts.

$NaOH + HCl \rightarrow NaCl + H_2O$

Uses:

It is used in the manufacture of soap, paper, artificial silk and a number of chemicals,

1. In petroleum refining.

2. In the purification of bauxite.
3. In the textile industries for mercerising cotton fabrics.
4. For the preparation of pure fats and oils.
5. As a laboratory reagent.

4. Sodium Carbonate (Na_2CO_3)

Preparation:

$NH_3 + H_2O + CO_2 \rightarrow NH_4HCO_3$

$NaCl + NH_4HCO_3 \rightarrow NaHCO_3 + NH_4Cl$

$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$

Properties:

1. Sodium carbonate is a white crystalline solid.
2. $Na_2CO_3.10H_2O$ is known as washing soda.
3. Sodium carbonate reacts with acids to give carbon dioxide.

Uses:

1. It is used in water softening, laundering and cleaning.
2. It is used in the manufacture of glass, soap, borax and caustic soda.
3. It is used in paper, paints and textile industries.
4. It is an important laboratory reagent both in qualitative and quantitative analysis.

$Na_2CO_3 + HCl \rightarrow NaCl + H_2O + CO_2$

Properties:

On heating sodium bicarbonate loses CO_2 and H_2O forming Na_2CO_3.

$$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$$

5. Sodium Chloride (NaCl)

Manufacture of sodium chloride is done from sea water. Sea water is allowed to dry up under summer heat in small tanks and solid crust so formed is collected.

Properties:

1. Sodium chloride is a white crystalline solid.
2. It is slightly hygroscopic.
3. It is soluble in water and insoluble in alcohol.

Uses:

1. It is used as a common salt or table salt for domestic purpose.
2. It is used for the preparation of Na_2O_2, NaOH and Na_2CO_3.

6. Sodium Bicarbonate (Baking Soda) ($NaHCO_3$)

Preparation:

When NaOH is treated with CO_2 in presence of H_2O it gives sodium bicarbonate.

$NaOH + CO_2 + H_2O \rightarrow NaHCO_3$

Properties:

On heating sodium bicarbonate loses CO_2 and H_2O forming Na_2CO_3.

$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$

Group-2 Elements: Alkaline Earth Metals

The elements of group-2 are Beryllium (Be), Magnesium (Mg), Calcium (Ca), Strontium (Sr), Barium (Ba) and Radium (Ra). Except for Be, rest are known as alkaline earth metals, because they were alkaline in nature and existed in the earth.

1. Electronic Configuration

The alkaline earth metals have 2 electrons in the s-orbital of the valence shell. Their general electronic configuration [Noble gas]ns^2

Element	Symbol	Electronic configuration
Beryllium	Be	$1s^22s^2$
Magnesium	Mg	$1s^22s^22p^63s^2$
Calcium	Ca	$1s^22s^22p^63s^23p^64s^2$
Strontium	Sr	$1s^22s^22p^63s^23p^63d^{10}4s^24p^65s^2$
Barium	Ba	$1s^22s^22p^63s^23p^63d^{10}4s^24p^64d^{10}5s^25p^66s^2$ or $[Xe]6s^2$
Radium	Ra	$[Rn]7s^2$

2. Atomic and Ionic Radii

The atomic radii as well as ionic radii of the members of the family are smaller than the corresponding members of alkali metals. Within the group, the atomic and ionic radii increase with increase in atomic number.

3. Ionization Enthalpies

The alkaline earth metals owing to their large size of atoms have fairly low values of ionization enthalpies. Within the group, the ionization enthalpy decreases as the atomic number increases.

4. Hydration Enthalpies

The hydration enthalpies of alkaline earth metal ions are larger than those of alkali metal ions. Therefore, compounds of alkaline earth metals are more extensively hydrated, for example, magnesium chloride and calcium chloride exist. the hydration enthalpies of alkaline earth metal ions decrease with increase in ionic size down the group. $Be^{2+} > Mg^{2+} > Ca^{2+} > Sr^{2+} > Ba^{2+}$

5. Physical Properties

The alkaline earth metals are silvery white, lustrous and relatively soft but harder than the alkali metals. The melting and boiling points of these metals are higher than the corresponding alkali metals. The electropositive character increases down the group from Be to Ba, Calcium, strontium and barium impart characteristic brick red, crimson and apple green colours respectively to the flame. The alkaline earth metals just like those of alkali metals have high electrical and thermal conductivities.

6. Chemical Properties

As compared to alkali metals, alkaline earth metals are less reactive due to their relatively higher ionization enthalpies. The reactivity of alkaline earth metals increases on going down the group.

i. Reaction with water: Ca Sr, and Ba have reduction potentials similar to those of corresponding group Ist metals and are quite high in the electrochemical series. They react with cold water readily, liberating hydrogen forming metal hydroxides.

$$Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$$

ii. Reaction with Air: Except Be these metals are easily tarnished in air as a layer of oxide is formed on their surface. Ba in powdered form bursts into flame on exposure to air.

iii. Reaction with hydrogen: The elements Mg, Ca, Sr and Ba all react with hydrogen to form hydrides MH_2.

iv. Reaction with oxygen: Except Ba and Ra the elements when burnt in oxygen form oxides of the type MO.

v. Reaction with halogens: When heated with halogens the alkaline earth metals directly combine with them and form the halides of the type MX_2.

$$Ca + Cl_2 \rightarrow CaCl_2$$

vi. Reaction with acids: The alkaline earth metals readily react with acids liberating dihydrogen.

$$M + 2HCl \rightarrow MCl_2 + H_2$$

Uses of alkaline earth metals:

1. Beryllium is used in the manufacture of alloys. Cooper-Beryllium alloys are used in the making of high strength springs.
2. Metallic beryllium is used for making windows of X-rays tubes.
3. Magnesium, being a light metal, forms many light alloys with aluminum, zinc, manganese and tin.
4. Magnesium is used in flash powders and bulbs, incendiary bombs and signals.
5. Magnesium-aluminium alloys are used in aircraft construction.
6. Magnesium is used as sacrificial anode for the prevention of corrosion of iron.
7. A suspension of magnesium hydroxide in water (called milk of magnesia) is used as an ant-acid to control excess acidity in stomach.

8. Magnesium carbonate is an ingredient of tooth-paste.
9. Calcium is used in the extraction of metals from oxides which are difficult to reduce with carbon.
10. Calcium and barium metals are used to remove air from vacuum tubes, due to their tendency to react with oxygen and nitrogen at high temperature.
11. Radium salts are used for radio therapy of cancer.

Anomalous Behaviour of Beryllium

Beryllium shows different behaviour from the rest members of its group and shows diagonal relationship to aluminium due to reasons discussed below.

1. Beryllium has exceptionally small atomic and ionic sizes and therefore does not compare well with other members of the group, because of high ionisation enthalpy and small size it forms compounds which are largely covalent and get easily hydrolysed.
2. Beryllium does not exhibit coordination number more than four as in its valence shell, there are only four orbitals. The remaining members of the group can have a coordination number of six by making use of d-orbitals.
3. The oxides and hydroxide of beryllium unlike the hydroxide of other elements in the group, are amphoteric in nature.

Compounds of Calcium

1. Calcium Oxide (CaO)

Preparation:

Calcium carbonate when decomposed at 800°C gives calcium oxide.

$$CaCO_3 \rightarrow CaO + CO_2$$

Properties:

1. Calcium oxide is also known as 'Quick lime' or 'Burnt lime', is white amorphous substance.
2. When water is added to lime a hissing sound is produced along with clouds of steam. The lime forms slaked lime [$Ca(OH)_2$.
3. Calcium oxide reacts with moist chlorine to form bleaching powder.

$$CaO + Cl_2 \rightarrow CaOCl_2$$

4. Calcium oxide on reaction with moist HCl gas forms $CaCl_2$.

$$CaO + 2HCl \rightarrow CaCl_2 + H_2O$$

2. Calcium Carbonate ($CaCO_3$)

Preparation:

Carbon dioxide when passed through lime water gives calcium carbonate.

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$$

Properties:

1. Calcium carbonate is a white powder insoluble in water.
2. Calcium carbonate dissolves in water in presence of CO_2 due to formation of calcium bicarbonate.

$$CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$$

3. Calcium Chloride ($CaCl_2$)

Preparation:

Calcium oxide, calcium hydroxide or calcium carbonate when treated with HCl gives calcium chloride.

$$CaO + 2HCl \rightarrow CaCl_2 + H_2O$$

Properties:

1. Calcium chloride is a colourless deliquescent crystalline substance which is soluble in water as well as in alcohol.
2. Crystals of calcium chloride when strongly heated gives off water of crystallizations.

4. Calcium sulphate (Plaster of Paris)

Preparation:

When Gypsum is heated at about 120° – 130°C, Plaster of Paris is formed.

$2CaSO_4 + 4H_2O \rightarrow (CaSO_4)2H_2O + 3H_2O$

Properties:

1. It is a white crystalline solid. It is sparingly soluble in water.
2. It becomes anhydrous at about 200°C. Anhydrous form is known as dead burnt plaster.

5. Calcium hydroxide $Ca(OH)_2$

Preparation:

$$CaO + H_2O \rightarrow Ca(OH)_2$$

Properties:

1. It gives $CaCO_3$ and $Ca(HCO_3)_2$ with CO_2

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$$

2. On prolong treatment with CO_2 milkiness disappears due to formation of $Ca(HCO_3)_2$

$$CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO3)_2$$

Diagonal Relationship of Lithium And Magnesium

Diagonal Relationship of Lithium with Magnesium

The main points of similarity:

(1) Both have almost similar electronegativities.

(2) Both Li and Mg are quite hard. They are harder and lighter than other elements in their respective groups.

(3) Both LiOH and $Mg(OH)_2$ are weak bases.

(4) Both form ionic nitrides when heated in atmosphere of Nitrogen.

$$6\ Li + N_2 \rightarrow 2\ Li_3N$$

$$3\ Mg + N_2 \rightarrow 2\ Mg_3N_2$$

(5) The hydroxides of both lithium and magnesium decompose upon heating.

$$2\ LiOH \rightarrow Li_2O + H_2O$$

$$Mg(OH)_2 \rightarrow MgO + H_2O$$

(6) Both lithium and magnesium combine with oxygen to form monooxide while other members of their respective groups form peroxide and superoxide.

$$4\ Li + O_2 \rightarrow 2\ Li_2O$$

$$2\ Mg + O_2 \rightarrow 2\ MgO$$

The Li_2O and MgO thus formed do not combine with excess O_2 to form peroxide and superoxide.

(7) The carbonates of these metals decompose on heating to the corresponding oxides with the evolution of carbon dioxide.

$$Li_2CO_3 \rightarrow Li_2O + CO_2$$

$$MgCO_3 \rightarrow MgO + CO_2$$

(8) Both Lithium and magnesium do not form solid bicarbonates.

(9) Both Lithium and magnesium nitrate decompose on heating producing nitrogen dioxide.

$$4\ LiNO_3 \rightarrow 2\ Li_2O + 4\ NO_2 + O_2$$

$$2\ Mg\ (NO3)\ 2 \rightarrow MgO + 4\ NO_2 + O_2$$

(10) The hydroxides, carbonate, oxalates, phosphates and fluorides of both Lithium and magnesium are sparingly soluble in water.

(11) Because of the covalent character, LiCl and $MgCl_2$ are soluble in ethanol.

(12) Both Lithium perchlorate and magnesium perchlorate are highly soluble in ethanol.

(13) LiCl and $MgCl_2$ are deliquescent and crystallise from aqueous solution as hydrate $LiCl_2\ H_2O$ and $MgCl_2\ 6\ H_2O$.

Extraction of Alkali Metals

Alkali metals cannot be extracted from their ores by the usual methods of extraction of metals because of following difficulties.

1) Alkali metals are strong reducing agents and hence cannot be extracted by reduction of their oxides or chlorides.

2) Alkali metals being highly electropositive cannot be displaced from the aqueous solution of their salts by other metals.

3) Alkali metals cannot be isolated by electrolysis of the aqueous solution of their salts since hydrogen is liberated at the cathode instead of the alkali metal because the standard electrode potentials of alkali metals are much lower than that of water. By using mercury as cathode, the alkali metals can be deposited at the cathode but the alkali metals so deposited readily combines with mercury to form an amalgam from which its recovery is very difficult.

Extraction of Lithium

Lithium is prepared by electrolysis of fused mixture of dry lithium chloride and potassium chloride at 723 k.

Potassium chloride is added to increase the conductivity of lithium chloride and also to lower the fusion temperature. The cell is operated at a temperature of about 723 K and current voltage of 8-9 V.

As a result of electrolysis, the following reactions takes place:

$$LiCl \Leftrightarrow Li^+ + Cl^-$$

At cathode: $Li^+ + e^- \rightarrow Li$

At anode: $2\ Cl^- - 2e^- \rightarrow Cl_2$

The metal thus obtained is 99% pure and is stored by keeping it wrapped in paraffin wax. Lithium being the lightest metal known cannot be stored in kerosene oil since it floats on the surface.

Uses of Lithium

1) It is used in the manufacture of alloys

a) Lithium – lead alloy or white metal which is used for making toughened bearing for motor engines and sheats for cables.

b) Lithium aluminium alloy has great tensile strength and electricity like that of mild steel. It is used for aircraft construction.

c) Lithium magnesium alloy is extremely tough and corrosion resistant which is used for armour plate and aerospace components.

2) It is used for producing thermonuclear energy required for propelling rockets and guided missiles.

3) Lithium is used to make both primary and secondary batteries.

4) Lithium is used as a Scavenger since it combines readily with oxygen and nitrogen. Thus it is used for removing little traces of oxygen and nitrogen during refining of metals.

5) lithium carbonate is used in making a special variety of glass which is very strong and is weather proof.

6) Lithium chloride is used in Air conditioning plants to regulate the humidity. It is also used in Ni-Fe accumulators.

7) Lithium Bromide is used in medicine as sedative.

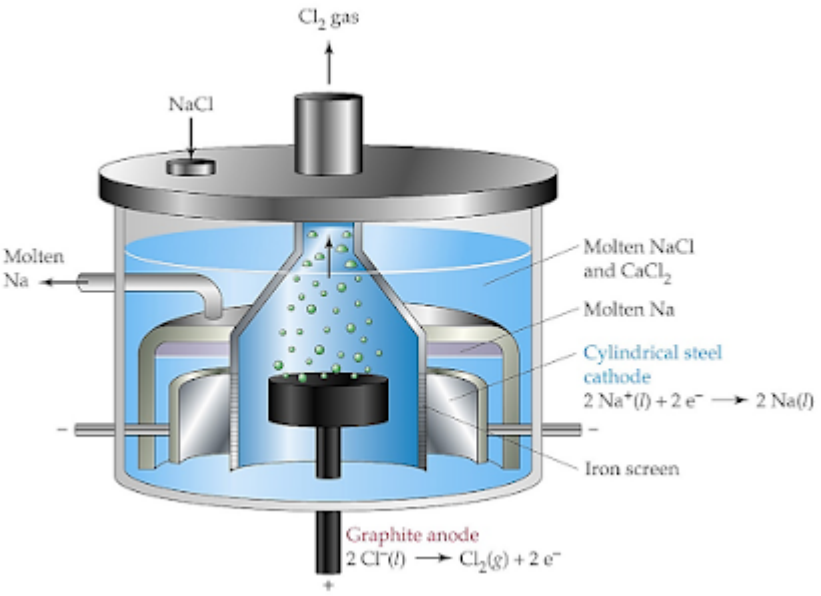
8) Lithium bicarbonate and lithium salicylate have been used for treatment of rheumatism since the resulting Lithium urinate is soluble in water.

9) Lithium hydride is used as source of hydrogen for meterological purpose and for filling of balloons.

10) Lithium Hydroxide is used for removing carbon dioxide from exhaled air in closed quarters.

11) Lithium aluminium hydride is used as a reducing agent in synthetic organic chemistry.

Extraction of Sodium



Sodium is extracted by the electrolysis of fused sodium chloride by a process called Down's process.

Sodium is now obtained by electrolysis of a fused eutectic mixture of sodium chloride and calcium chloride in Down's cell at 873 K using graphite anode and iron cathodes.

As a result of electrolysis, sodium is liberated at the cathode and Cl_2 is evolved at the anode.

$$NaCl \rightarrow Na^+ + Cl^-$$

$$\text{At cathode: } Na^+ + e^- \rightarrow Na$$

$$\text{At anode: } Cl^- \rightarrow Cl + e^-$$

$$Cl + Cl \rightarrow Cl_2$$

Uses of Sodium

1) Sodium is used as a reducing agent in the extraction of boron and silicon.

2) Sodium is employed as a reducing agent in form of sodium amalgam and as a reagent in Wurtz reaction and in the synthesis of many organic compound.

3) It is also used to make tetraethyl lead,tetra methyl lead which are used as anti knocking agent for gasoline.

4) liquid Na or its alloys with potassium is used as a coolant in fast breeder nuclear reactor.

5) It is used in the manufacture of number of chemicals such as Na_2O_2, NaCN, $NaNH_2$.

6) It is used in sodium vapour lamp.

7) Sodium is largely used in industry for the production of artificial rubber, dyes, drugs.

8) Because of its lightness and high thermal conductivity, it is used for filling exhaust valves of aeroplane engines.

Use of Potassium

1) It has a vital role in biological system.

2) KCl is used as fertilizer.

3) KOH is used in manufacture of soap.

4) It is also used as an excellent absorbent for Carbon dioxide.

Baking Soda and Sodium Chloride

Sodium chloride (NaCl)

The most abundant source of sodium chloride is sea water which contains 2.7 – 2.9% by mass.

Common salt is generally obtained by evaporation of sea water. Crude sodium chloride obtained by crystallisation of brine solution contains impurities of sodium sulphate (Na_2SO_4), calcium sulphate ($CaSO_4$), calcium chloride ($CaCl_2$) magnesium chloride ($MgCl_2$).

Since $MgCl_2$ and $CaCl_2$ are deliquescent (absorb moisture easily from the atmosphere), therefore, impure common salt gets wet in rainy season. To obtain pure sodium chloride , the crude salt is dissolved in minimum amount of water and filtered to remove insoluble impurities. The solution is then saturated with HCl gas when crystals of pure sodium chloride separates out. The solution is then saturated with hydrogen chloride gas when crystals of pure sodium chloride separates out. The calcium and magnesium chloride being more soluble remains in the solution.

Properties

1) Sodium chloride melts at 1081 K.

2) Its solubility is 3.6 g per 100 g of H_2O at 273 K. The solubility does not increase appreciably with rise in temperature.

Uses

1) It is used as a common salt for domestic purpose.

2) It is used in the preparation of Na_2CO_3, NaOH, Na_2O_2.

Baking soda or Sodium hydrogen carbonate ($NaHCO_3$)

Sodium hydrogen carbonate is commonly called Baking soda because on heating it decomposes to evolve bubbles of carbon dioxide.

Preparation

It is prepared by saturating a solution of sodium carbonate with carbon dioxide. Being less soluble, white crystalline powder of sodium hydrogen carbonate gets separated.

$$Na_2CO_3 + CO_2 + H_2O \rightarrow 2\ NaHCO_3$$

Uses

1) It is used in fire extinguishers for generating CO_2.

$$NaHCO_3 + HCl \rightarrow NaCl + CO_2 + H_2O$$

2) It is used in the preparation of baking powder which is a mixture of $NaHCO_3$, starch, calcium dihydrogen phosphate $Ca(H_2PO_4)$, sodium Aluminium Sulphate $NaAl\ (SO_4)_2$.

3) It is a mild antiseptic for skin infections.
4) It is used as an antacid in making digestive powders for removing acidity of stomach.

Cement

Cement is essentially a finely powdered mixture of calcium silicate and aluminates along with small quantities of gypsum which sets into a hard stone like mass when treated with water.

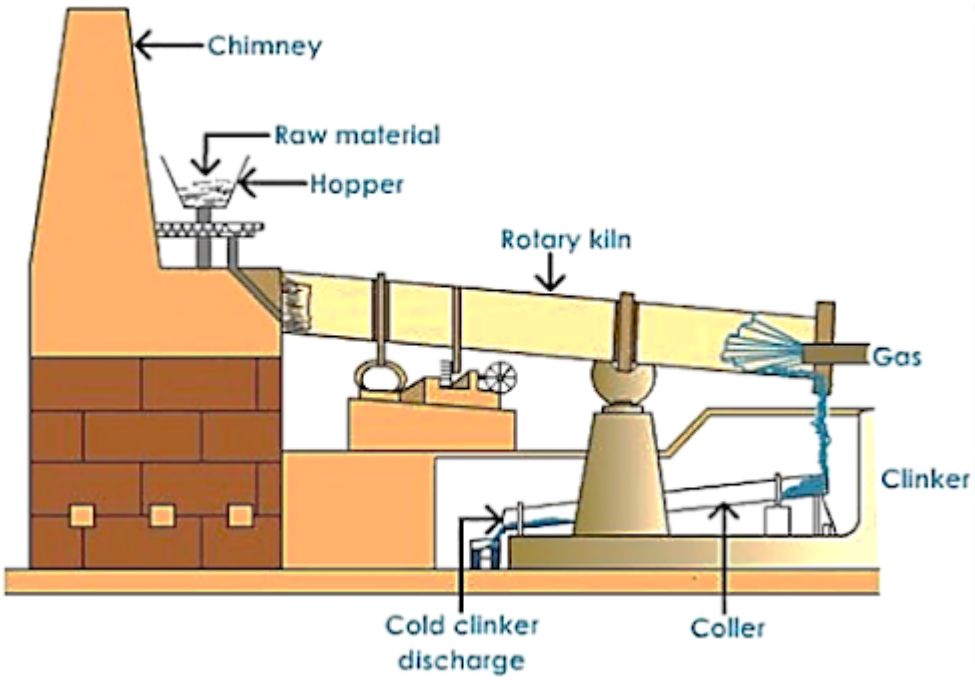
Average composition of Portland cement

Cement is obtained by combining a material rich in lime, CaO with other materials such as clay, which contains silica, SiO_2 along with oxides of aluminium, iron and magnesium.
Lime (CaO) 50-60%
Magnesium oxide (MgO) 2-3%
Silica (SiO_2) 20-25%
Ferric oxide (Fe_2O_3) 1-2%
Alumina (Al_2O_3) 5 – 10%
Sulphur trioxide (SO_3) 1-2%
For a good quality cement, the ratio of alumina to silica should lie between 2.5 to 4 while that of lime to that of silica,alumina and ferric oxide should be as close to 2 as possible.

Raw material

Raw materials required for the manufacture of cement are 1) limestone ($CaCO_3$), 2) clay which provides both silica and Alumina 3) gypsum. Small amount of magnesia and iron oxide are also required for imparting suitable colour to cement.

Manufacture of cement



Cement is manufactured by two processes:

(1) Wet process

(2) Dry process

The wet process is preferred when the raw materials i.e. limestone and clay are soft, climate is humid and fuel is cheap. The dry process is preferred when the raw materials are hard.

The main raw materials are limestone and clay. Wet process supplies lime while the dry process is a source of silica, alumina and iron oxides.

(1) The raw materials are first crushed separately in a suitable machine. They are then mixed in the required proportion (3 parts limestone + 1 part clay) and grounded together finely. This grinding is done either by the dry process or by the wet process.

(2) In the wet process, the clay is washed with water in a wash mill to remove flint and other foreign substances. It is then mixed with requisite quantity of limestone and pulverised in a special mill. The resulting pasty mass is then throughly homogenised to get raw slurry containing about 40% water.

(3) In the dry process, the raw materials are dried and mixed in desired proportion. The mixture is then finely powdered and passed through 300 mesh sieves and homogenised with the help of compressed air. The homogeneous mass is called raw meal.

(4) The slurry or raw meal is introduced into the upper end of a rotary kiln by means of screw conveyer. The kiln consist of an inclined steel rotating cylinder, 150-200 ft long and about 10 ft diameter lined inside with fire bricks. The charge travel downwards slowly due to the rotatory motion given to the kiln and is heated by burning coal dust which is blown in front from the lower end. The charge takes about 2-3 hours to cover the entire journey in the kiln.

The following changes occur:

(1) In the upper part of the kiln, the temperature is around 1000-1100K .Here the charge loses all its water due to evaporation by hot gases.

(2) In the middle of the kiln, temperature is around 1100-1200 K. Here limestone decomposes to form calcium oxide and carbon dioxide.

$$CaCO_w \rightarrow CaO + CO_2$$

(3) The lower part of the kiln, from where coal dust is blown in is the hottest with the temperature ranging between 1770-1870 K. As the charge reaches here, chemical combination takes place between lime, alumina and silica to form calcium silicate and aluminates.

$$2\ CaO + SiO_2 \rightarrow 2\ CaO.SiO_2$$

$$3\ CaO + SiO_2 \rightarrow 3CaO.\ SiO_2$$

$$3\ CaO + Al_2O_3 \rightarrow 3\ CaO.Al_2O_3$$

$$2\ CaO + Al_2O_3 \rightarrow 2\ CaO.Al_2O_3$$

$$4\ CaO + Al_2O_3 + Fe_2O_3 \rightarrow 4\ CaO.\ Al_2O_3.Fe_2O_3$$

Due to very high temperature in this zone, about 20-30 % combines with solid mass to form grey coloured balls called cements clinkers.

The hot clinkers are cooled by cold air and are mixed with 2-3 % gypsum and finely powdered to 325 mesh size in grinding machine. The fine powder called the Portland cement is sieved and packed in bags.

Setting of cement

The important constituents present in Portland cement are dicalcium silicate (Ca_2SiO_4), tricalcium silicate (Ca_3SiO_5) and tricalcium aluminate ($Ca_3Al_2O_6$).
When water is added to cement, an exothermic reaction occurs. During this process, cement reacts with water to form a gelatinous mass which slowly sets into a hard mass. having three dimensional network structure involving -Si-O-Si- and -Si-O-Al chains.
Tricalcium silicate sets quickly and develops considerable strength within a few days. Dicalcium silicate sets slowly and develops appreciable strength after a month or so. Tricalcium aluminate sets instantaneously in presence of water. The internal strength acquired by cement is primarily due to the setting of tricalcium aluminate.

Summary-

1. s-block elements: The elements in which last electron enters into s-orbital are called s- block elements.
2. Alkali metals: The elements of group 1 whose hydroxide are strong alkali.
3. Alkaline earth metal: The elements of group 2, and their oxides and hydroxides are alkaline in nature and their oxides are found in the Earth's crust.
4. Diagonal relationship: The resemblance in properties of elements of second period with elements of third period present diagonally on the right hand side.
5. Monovalent sodium and potassium ions and divalent magnesium and calcium ions are found in large proportions in biological fluids. These ions perform important biological functions such as maintenance of ion balance and nerve impulse conduction.

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